

From Aid to Autonomy

A World Bank IDA Lending Analysis of South Asia & Bangladesh

SQL (MySQL) + Tableau Data Analytics Project

1.4M+ rows | IDA Statement of Credits, Grants & Guarantees

7 South Asian Countries | 1,079 Projects | \$94.9B Loaned

01 Project Overview

This project analyzes decades of World Bank **International Development Association (IDA)** lending data to understand how South Asian nations — and Bangladesh in particular — borrow, disburse, and repay development financing. Across the region, the IDA has loaned out a staggering **\$94.9 billion**, with **\$44.3 billion** still outstanding, spread across **1,079 projects** in **7 countries**.

The analysis moves from a regional lens — comparing the top borrowing countries in South Asia on loan volume, outstanding debt, and repayment performance — down to a country-level deep dive on **Bangladesh**, one of the most active borrowers in the dataset. Using SQL for data cleaning, de-duplication, and aggregation, and Tableau for visualization, the project traces Bangladesh's borrowing cycles, disbursement efficiency, repayment pace, and the evolving mix of credits, grants, and guarantees that make up its financing relationship with the World Bank.

Beyond the numbers, the project asks a bigger question: *what can a country's loan data teach us about its progress, resilience, and reputation on the global stage?* The findings position Bangladesh as an active, high-capacity borrower that is steadily transitioning from short-term, aid-dependent financing toward larger, long-term development partnerships.

02 What Sparked This Project?

This project began as an assignment in a data analysis bootcamp, where the brief was simple on the surface: use SQL to analyze a large dataset and extract meaningful insights. With dozens of countries in the dataset and no obvious starting point, one country's data stood out for its depth and activity spanning decades of borrowing, disbursement, and repayment: **Bangladesh**.

In recent years, Bangladesh has emerged as a notable example of steady, determined growth. Despite facing significant challenges — from natural disasters to climate pressures — the country has achieved impressive gains in economic development, poverty reduction, and human well-being. Between 2010 and 2023, GDP growth averaged **6.4%**, while the poverty rate dropped from **11.8% to 5.0%**. These milestones suggested a country reshaping its future through consistent effort and strategic support — and a compelling story hiding inside a loan ledger.

What started as a technical SQL exercise became something more personal: a chance to practice querying and data storytelling while pivoting into a new career in data — and a reminder that behind every dataset is a real-world narrative of progress.

03 Tools & Technologies

Tool	Role in the Project
SQL (MySQL)	Data cleaning, de-duplication, transformation, and aggregation. Used CTEs, window functions, and subqueries to build clean, analysis-ready subsets of the data.

Tableau	Turned SQL query outputs into line charts and stacked bar charts to visualize borrowing, repayment, disbursement, and funding-mix trends over time.
SQL Aliasing	Renamed technical World Bank column headers into clear, human-readable labels for easier analysis and reporting.

04 Dataset

The dataset is the World Bank's **IDA Statement of Credits, Grants, and Guarantees**, containing **1.4 million+** rows and **30+ columns** of historical lending data, converted into USD.

Scope	Detail
Regional (South Asia)	7 countries, 1,079 projects, \$94.9B loaned, \$44.3B outstanding
Bangladesh Deep-Dive	Filtered to Bangladesh-only records (~52,000–68,000 rows after filtering), covering borrowing, disbursement, repayment, and credit-type columns
Project-Level View	50 IDA-funded projects categorized into six groups: Finance, Energy, Clean Air, Health, Transportation, and Other
Key Columns Used	Loan/Credit ID, board approval date, committed amount, disbursed amount, repaid amount, outstanding balance, service charge rate, credit-type prefix

Top 3 South Asian Borrowing Countries (Snapshot)

Country	Total Disbursed	Share of Region	% Repaid
India	\$46.30B	48.7%	74.27%
Bangladesh	\$33.24B	— (2nd largest)	2nd highest repayment
Nepal	\$7.75B	—	—

Note: Bangladesh ranks 2nd in total loan amount, 1st in outstanding debt, and 2nd in repayments made among South Asian IDA borrowers.

05 SQL Analysis (MySQL)

Performed **11 SQL queries** ranging from basic aggregations to advanced **window functions**, **CTEs (Common Table Expressions)**, and **subqueries**, covering everything from regional country rankings down to year-by-year Bangladesh borrowing behavior.

- **Regional ranking queries** — aggregated loan, disbursement, and repayment totals by country to identify the top 3 South Asian borrowers and rank repayment responsibility.

- **De-duplication CTEs** — used CTEs with MIN/MAX aggregation on unique loan IDs to ensure each loan was counted once despite appearing across multiple repayment/disbursement rows.
- **Year-over-year aggregation** — grouped committed, disbursed, and repaid amounts by first board-approval year to build borrowing, repayment, and disbursement-gap trend queries.
- **CASE-based classification** — categorized loans as credit, grant, or guarantee using World Bank ID-prefix logic, and grouped 50 projects into six purpose-based categories (finance, energy, clean air, health, transportation, other).
- **Ratio & percentage calculations** — computed disbursement rates, repayment percentages, average service charge rates, and loan exposure as a share of GDP.
- **Column aliasing & cleanup** — renamed raw World Bank headers into clear, analysis-friendly labels across all queries.

06 Business Questions & Answers

The core of this project is a set of business questions that move from a regional comparison of South Asian IDA borrowers to a detailed, year-by-year profile of Bangladesh's borrowing behavior. Each question below is answered directly from the SQL query results and Tableau visualizations.

Part A — Regional Analysis: South Asia

Q1. Which countries are the top 3 IDA borrowers in South Asia, and how much do they still owe?

Method: Regional SQL aggregation grouping total borrowed, disbursed, and outstanding amounts by country.

Answer:

India is the largest recipient of IDA financing, with a total disbursed amount of over **\$46.30 billion** — accounting for **48.7%** of all funds loaned by the IDA in South Asia. Bangladesh follows closely with **\$33.24 billion** disbursed, and Nepal ranks third at **\$7.75 billion**. Together, these three countries represent the bulk of the region's \$94.9B in total IDA lending across 1,079 projects.

Q2. Which South Asian countries have been the most responsible in repaying their loans to the IDA?

Method: Ranking query comparing % of borrowed amount repaid, outstanding balance, and total loan volume by country.

Answer:

India is the most disciplined repayer among the top borrowers, having repaid **74.27%** of what it has borrowed — the highest repayment percentage of the group. **Bangladesh** ranks **2nd** in repayments made, but interestingly also carries the **largest outstanding debt balance** in the region despite being only the 2nd-largest borrower by total amount — a signal of larger, longer-duration financing rather than weaker repayment discipline.

Q3. How does Bangladesh's cost of borrowing compare to the South Asia regional average?

Method: Average() aggregation of service charge rate, grouped by country vs. regional average.

Answer:

Bangladesh's average IDA service charge rate is **0.82%**, notably lower than the South Asia regional average of **1%**. This lower average service charge means Bangladesh's financing terms are, on average, more favorable than the regional norm — a modest but meaningful advantage on billions of dollars in outstanding financing.

Q4. How are the World Bank's IDA-funded projects distributed by purpose?

Method: CASE-based classification of project purpose combined with a repayment-percentage calculation per project.

Answer:

Across 50 IDA-funded projects analyzed, financing was categorized into six primary purpose groups: **Finance, Energy, Clean Air, Health, Transportation, and Other**. For each project, the percentage of the borrowed amount already repaid was also calculated, allowing repayment performance to be compared not just by country but by the type of development work being funded.

Part B — Bangladesh Deep-Dive

Q5. Is Bangladesh a reliable borrower? How has its borrowing and repayment behavior changed over time?

Method: CTE-based de-duplication of loan records by unique loan ID, joined with year-over-year borrowing and repayment aggregation.

Answer:

Bangladesh shows the profile of an increasingly trusted, high-capacity borrower. Its borrowing occurs in distinct **cyclical waves**, and while early-era loans were repaid quickly (shorter terms, smaller amounts), repayments on more recent, larger loans lag behind — not because of weaker discipline, but because newer loans come with longer grace periods typical of large-scale development finance. Overall, Bangladesh continues to borrow at scale, repay actively, and disburse a high share of committed funds.

Q6. How much has Bangladesh borrowed over the years, and what patterns emerge?

Method: SQL grouping by first board-approval year using MIN (earliest approval date) and MAX (full borrowed amount) per unique loan ID, visualized as a Tableau line chart.

Answer:

Bangladesh's borrowing follows a **cyclical pattern** with noticeable peaks in **2011, 2018, and again in recent years**, likely coinciding with major infrastructure or development pushes. Between these peaks, borrowing stabilizes or dips — possibly reflecting shifts in policy priorities, repayment cycles, or the project pipeline.

Q7. Have Bangladesh's repayments kept pace with its recent borrowing?

Method: CTE-based de-duplication tracking borrowed vs. repaid amounts by year, visualized as a Tableau stacked bar chart.

Answer:

Not entirely — and that is not necessarily a red flag. In earlier decades, Bangladesh repaid loans quickly, reflecting shorter terms and smaller loan sizes. Since the 2010s, borrowing has surged, but repayments have not caught up at the same pace, largely because newer, larger loans include longer grace periods. This reflects Bangladesh's **transition from short-term borrowing to long-term development finance** as it invests in infrastructure, energy, and resilience.

Q8. Are Bangladesh's projects being executed as planned? What portion of committed funds has actually been disbursed?

Method: Year-by-year comparison of borrowed vs. disbursed amounts using a de-duplicating CTE, visualized as a Tableau stacked bar chart.

Answer:

Historically, Bangladesh has disbursed most of what it borrowed, a strong sign of implementation capacity. However, since **2020**, the gap between borrowed and disbursed amounts has widened: in 2023 and 2024, billions were committed but a much smaller share had been disbursed by the time of reporting, reflecting newer multi-year projects still ramping up. Bangladesh's overall disbursement rate stands at **67.55%** — the lowest in South Asia, yet still considered a very strong indicator internationally (rates near 70% suggest an active pipeline, not one that is bottlenecked or stalled).

Q9. Is Bangladesh moving from aid dependency toward financial autonomy? Has the funding mix shifted from grants to credits?

Method: CASE logic based on World Bank loan-ID prefix rules, grouped by year and credit type, visualized as a Tableau stacked bar chart.

Answer:

The funding mix has diversified over time, though not in the direction initially expected. Early on, Bangladesh relied almost entirely on **credits** (repayable financing). In recent decades, **grants** have grown modestly — with a noticeable uptick around **2020**, likely tied to emergency COVID-19 assistance — and **guarantees** remain a small but present share of the mix. Even with more grants entering the picture, Bangladesh continues to borrow at scale, repay actively, and disburse a high share of committed funds — the behavior of a trusted, high-capacity borrower rather than an aid-dependent one.

Q10. How large is Bangladesh's total IDA loan exposure relative to the size of its economy?

Method: Aggregation of total committed and disbursed loan amounts, divided by Bangladesh's national GDP figure.

Answer:

Bangladesh's total committed loans stand at **\$47.76 billion**, equal to roughly **10.47%** of the country's GDP. Of that, **\$33.24 billion** has actually been disbursed — about **7.29%** of GDP — representing real cash that has been delivered into the economy. This double-digit GDP exposure is meaningful and calls for continued, prudent debt-sustainability monitoring, but the high disbursement rate demonstrates Bangladesh's structural capacity to absorb large liquidity injections and convert them into productive, asset-building growth.

Q11. What was the most unexpected finding compared to initial assumptions?

Method: Comparative review of the credit/grant/guarantee mix query across early vs. recent time periods.

Answer:

Grants were expected to be more prominent in Bangladesh's earlier borrowing history, when the country was presumed to be more aid-dependent. The data showed the opposite: grants have only become more common in **recent** years. This is a reminder that development finance evolves alongside shifting global goals (such as climate resilience and inclusion), and that a country's support model can continue to shift even as its economy matures.

07 Key Business Insights

1

India Leads, and Repays, the Most

India is the largest IDA borrower in South Asia, holding 48.7% of the region's total lending (\$46.30B) and also posting the highest repayment rate (74.27%) among top borrowers — pairing scale with financial discipline.

2

Bangladesh Carries the Region's Largest Outstanding Debt

Despite ranking 2nd in total loan volume, Bangladesh holds the highest outstanding debt balance in South Asia — a signal of large-scale, long-duration financing rather than weak repayment behavior (it still ranks 2nd in repayments made).

3

Bangladesh Borrows on Favorable Terms

Bangladesh's average service charge rate (0.82%) is meaningfully below the South Asia regional average (1%), giving it a modest but real cost advantage on billions of dollars in financing.

4

Borrowing Moves in Predictable Cycles

Bangladesh's loan activity peaks around 2011, 2018, and in recent years — patterns that likely track major national infrastructure and development pushes rather than random, ad-hoc borrowing.

5

A Widening Disbursement Gap Signals Bigger, Slower Projects — Not Stalled Ones

Since 2020, disbursed amounts have increasingly lagged committed amounts. At 67.55%, Bangladesh's disbursement rate is the lowest in South Asia, yet still strong by international standards, reflecting newer multi-year projects still ramping up rather than a stalled pipeline.

6

The Funding Mix Is Diversifying, Not Shrinking

Bangladesh's financing has shifted from being almost entirely credit-based to gradually incorporating more grants (with a COVID-era spike around 2020) and a small share of guarantees, broadening its financial toolkit.

7

Loan Exposure Is Significant but Manageable

Committed loans equal 10.47% of GDP and disbursed loans equal 7.29% of GDP — a double-digit exposure that warrants ongoing debt-sustainability monitoring, balanced by strong evidence that Bangladesh can convert this capital into real, on-the-ground development outcomes.

Bottom line: Bangladesh's World Bank IDA data tells a story of steady, deliberate growth — moving from smaller, short-term loans toward larger, more complex, long-term financing, and from a nearly all-credit funding mix toward a more diverse toolkit that includes grants and guarantees. The combination of active borrowing, strong disbursement execution, and consistent repayment reflects the profile of a trusted, high-capacity

development partner.